

Cell wall inhibitors...continue

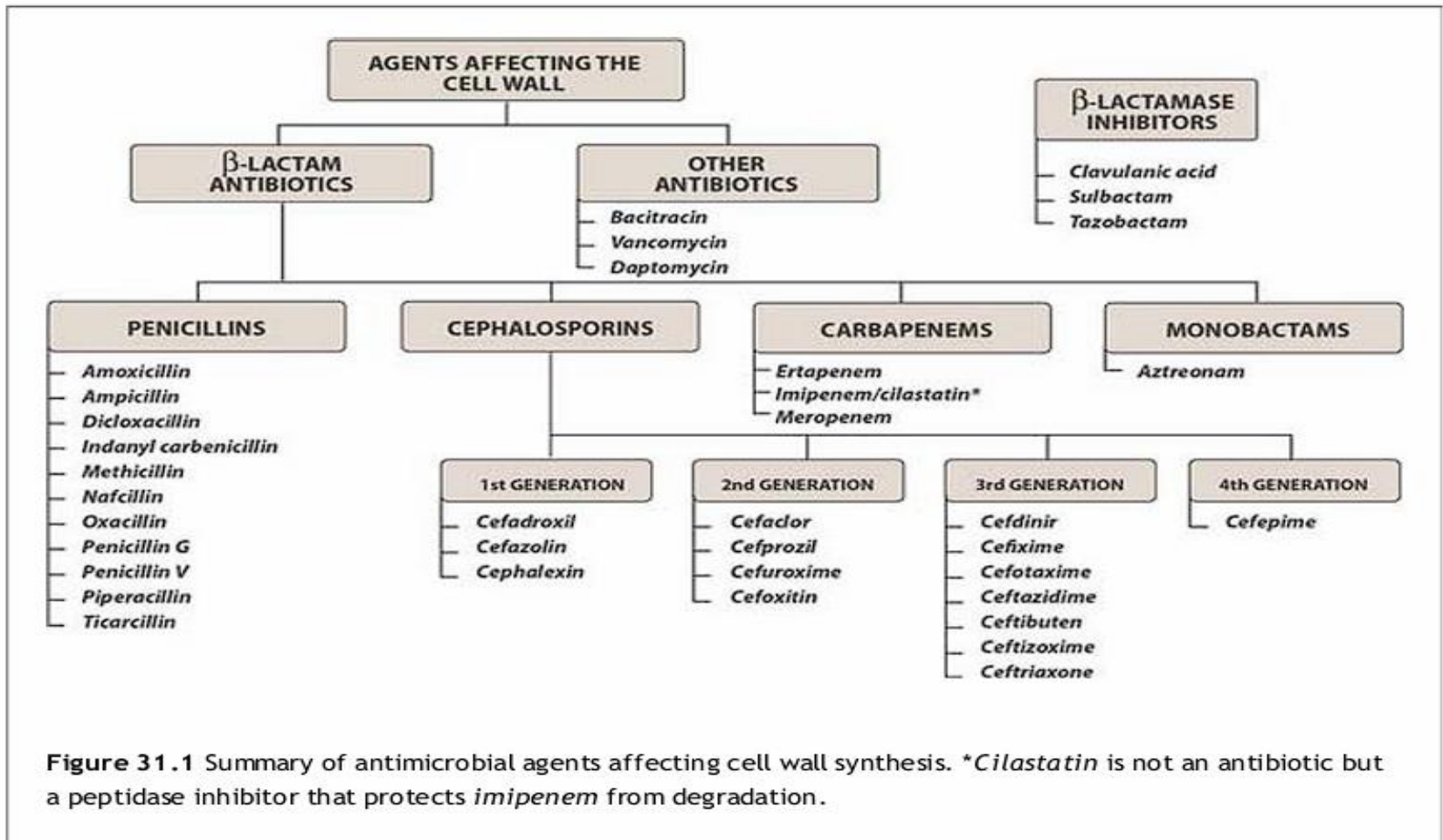


Figure 31.1 Summary of antimicrobial agents affecting cell wall synthesis. **Cilastatin* is not an antibiotic but a peptidase inhibitor that protects *imipenem* from degradation.

the cephalosporins

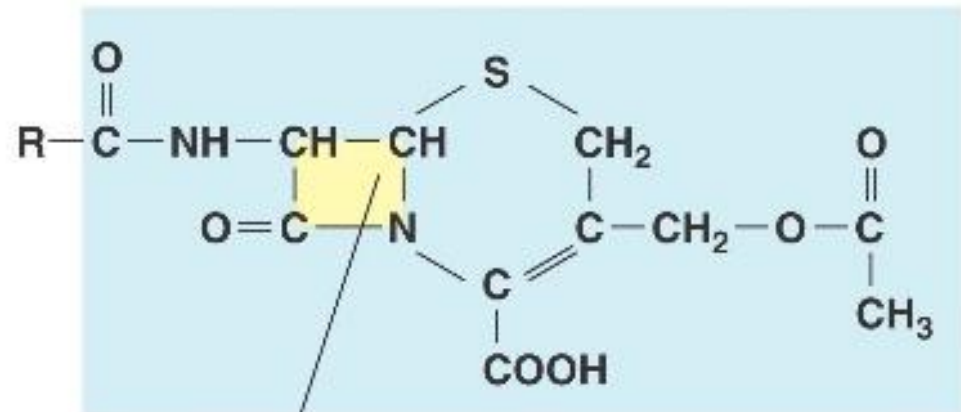
- Cephalosporins are β -lactam antimicrobials used to manage various infections caused by both gram-positive and gram-negative bacteria.
- The 5 generations of cephalosporins demonstrate efficacy in treating skin and soft tissue infections, pneumonia, meningitis, and other infections.
- Cefiderocol is a novel siderophore cephalosporin that exhibits remarkable antibacterial activity. This drug has been approved by the US Food and Drug Administration (FDA) to treat complicated urinary tract infections and ventilator-associated pneumonia caused by highly resistant gram-negative bacteria, including *Klebsiella pneumonia*, *Proteus mirabilis*, *Escherichia coli*, *Pseudomonas aeruginosa*, and *Acinetobacter baumannii*.

Cephalosporins *cont.*

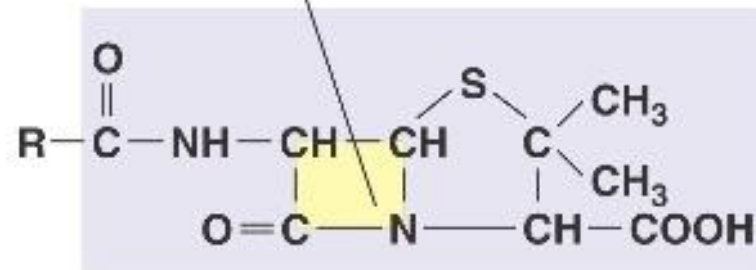
Structure and mode of action resembles penicilins

1. More stable to bacterial β -lactamases than penicilins

2. Broader spectrum
 $\Rightarrow$ used against penicillin-resistant strains



β -lactam ring Cephalosporin nucleus



Penicillin nucleus

Generations

– First generation cephalosporins

- cefadroxil
- Cefazolin
- Cephalexin
- Cephalotin
- Cephapirin
- Cephadrine

– Second Generation cephalosporins

- Cefaclor
- Cefamandole
- Cefonizind
- Cefotetan
- Cefoxitin
- Cefmetazole
- Cefprozil

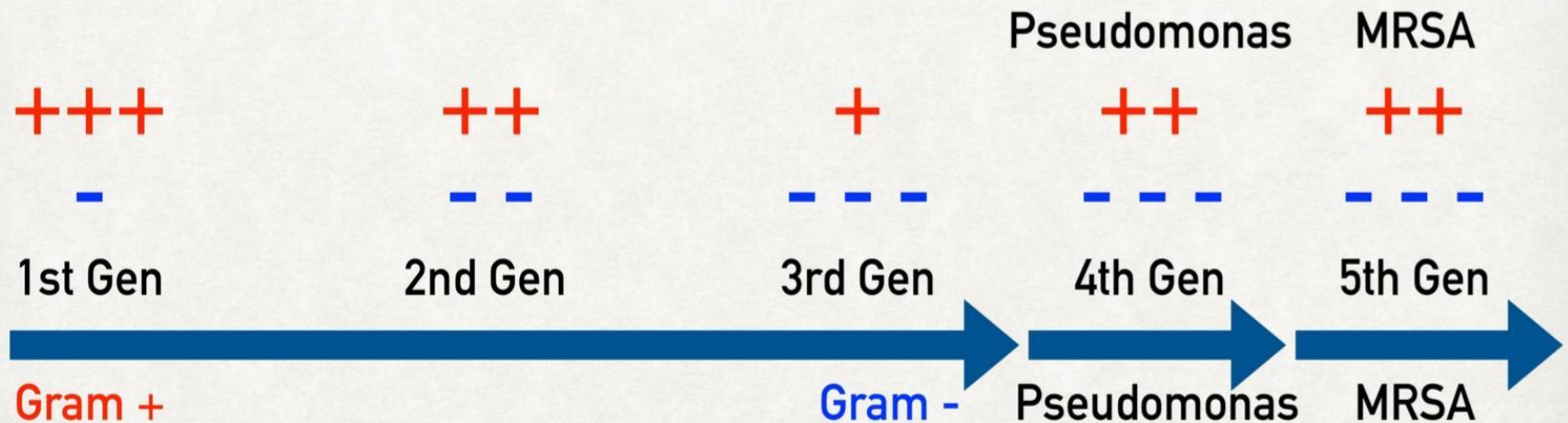
– Third Generation Cephaosporins

- Cefnidir
- Cefixime
- Cefoperazone
- Cefotaxime
- Cefpodoxime
- Ceftazidime
- Ceftibuten
- Moxalactam

– Fourth Generation Cephalosporin

- Cefepime

Cephalosporin Trick! Coverage Made Easy



THE CEPHALOSPORINS

- First Generation cephalosporins- are largely effective against the same gram-positive organisms affected by penicillin.
- Second generation cephalosporins- are effective against those strains as well as *Haemophilus influenza*, *Enterobacter aerogenes* and *Nisseria sp.* These drugs are less effective against gram positive bacteria

THE CEPHALOSPORINS

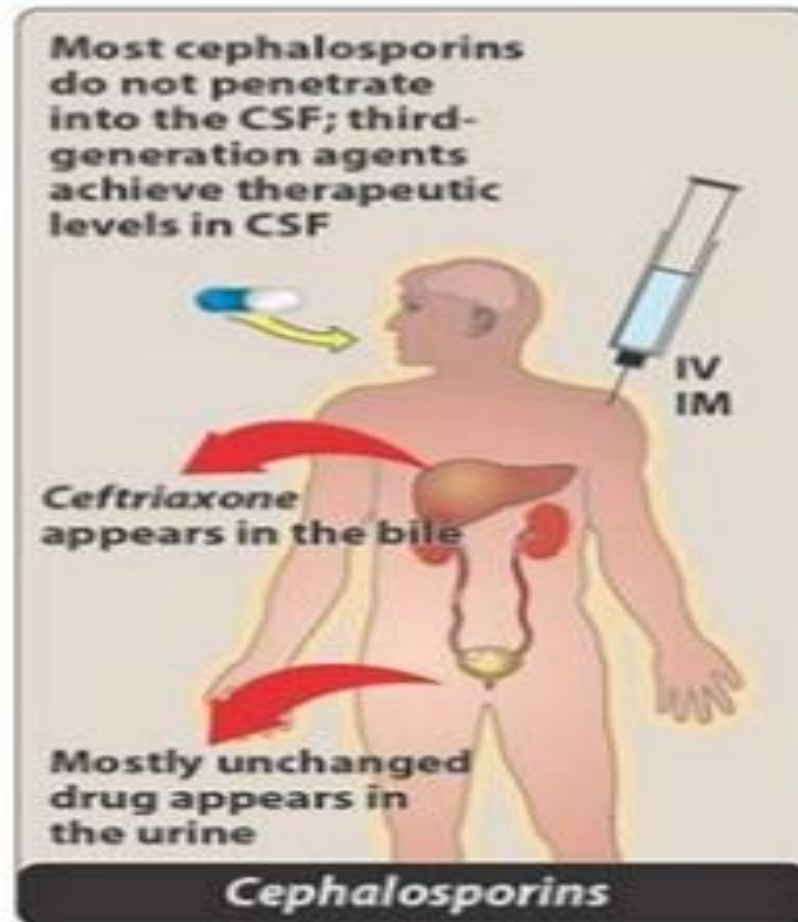
- Third Generation cephalosporins- are relatively weak against gram-positive bacteria but more potent against gram-negative bacteria, to include *Serratia marcescens*.
- Fourth generation cephalosporins- are developed to fight against the resistant gram-negative bacteria. The first drug is cefepime.

Cephalosporin

The mechanism of action:-

- The cephalosporins are primarily **bactericidal**.
- *They interfere with the cell-wall building ability of bacteria when they divide.*
- They prevent the bacteria from biosynthesizing the framework of their cell wall.
- The weakened cell wall will swell and burst causing cell death.

Administration and fate of the cephalosporins



Cephalosporin

- Pharmacokinetics
 - Only a few cephalosporins are administered orally, most are administered parenterally.
 - Their half-lives are short and they are **excreted mainly in the urine.**
- Contraindications and Precautions
 - *The drugs are contraindicated in patients with known allergies to cephalosporins and penicillins.*

Oral CS

First	Second	Third
Cephalexin	Cefatrizine	Cefixime
Cephadrine	Cefaclor	Ceftibuten
Cefadroxil	Cefroxadine	Cefdinir
	Cefproxil	Loracarbef
	Cefuroxime-axetil	Cefpodoxime-proxetil
		Cefetamet-pivoxil

Cephalosporin

Adverse Effects

- GI system- Nausea, vomiting, diarrhea, anorexia, abdominal pain and flatulence are common effects.
- CNS – headache, dizziness, lethargy and paresthesias have been reported.
- Renal system- nephrotoxicity in individuals with pre-existing renal disease
- Hypersensitivity

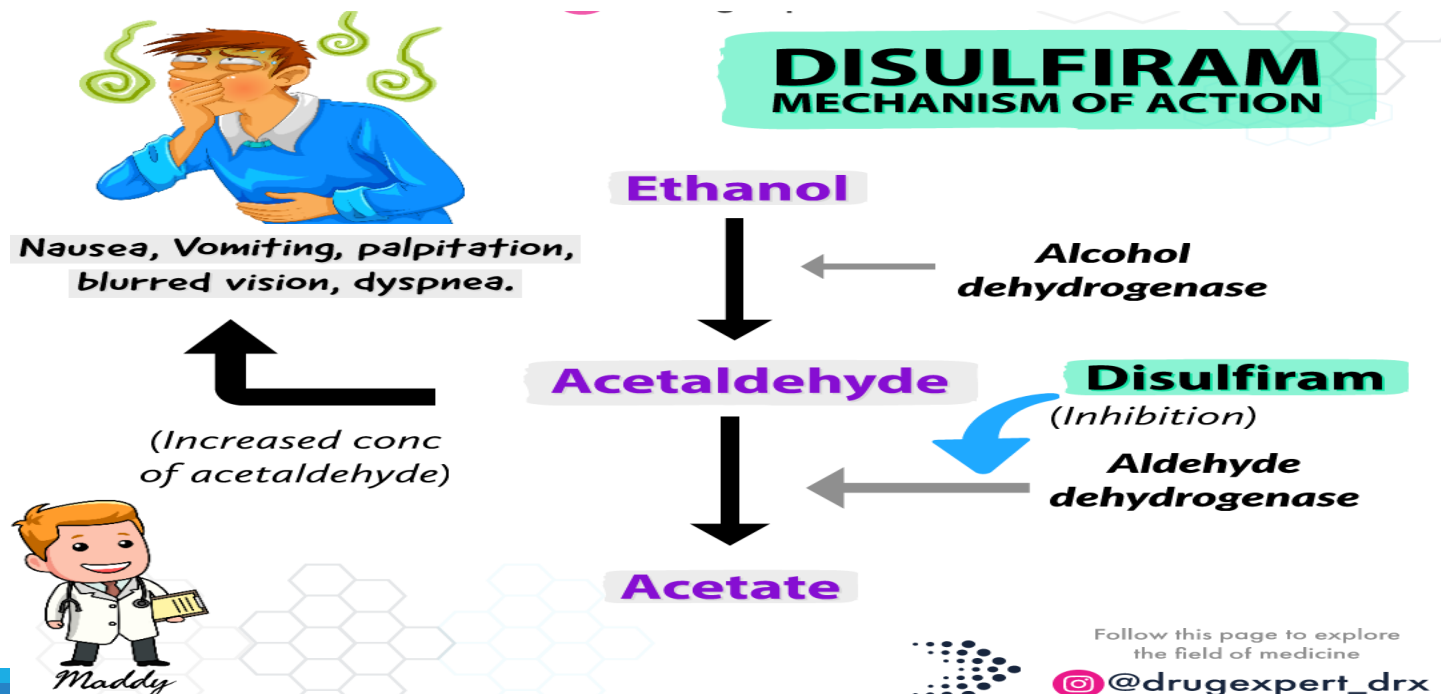
Cephalosporin

Drug-Drug interactions

- ALCOHOL- many patients experience a disulfiram-like reactions when taken with some specific cephalosporins (cefamandole, cefoperazone or moxalactam).
- The patient may experience flushing, headache, nausea, vomiting and muscular cramps. This may occur even up to 72 hours of cephalosporin discontinuance.

Disulfiram-like Reaction

- A **disulfiram-like reaction** happens when a drug blocks alcohol metabolism → leading to **acetaldehyde accumulation** after ethanol intake.
- It mimics the effect of Disulfiram (used for alcohol dependence).



Carbapenems

Carbapenems are a class of beta-lactam antibacterials that includes:

1.ertapenem

2.imipenem-cilastatin sodium (a combination drug)

3.meropenem.

- ✓ The antibacterial spectrum of activity for imipenem-cilastatin is broader than that of any other antibacterial studied to date.
- ✓ Because of this broad spectrum of activity, it's used for serious or life-threatening infection, especially gram-positive and gram-negative health-care acquired infections

Pharmacodynamics

- Imipenem-cilastatin, ertapenem, and meropenem are bactericidal.
- They exert antibacterial activity by inhibiting bacterial cell-wall synthesis.

Pharmacotherapeutics

Imipenem has the broadest spectrum of activity of currently available beta-lactam antibiotics:-

1. It's effective against aerobic gram-positive species, such as Streptococcus, S. aureus, and S. epidermidis.
2. It inhibits most Enterobacter species.
3. It also inhibits P. aeruginosa (including strains resistant to

Adverse reactions to carbapenems

Common adverse reactions to ertapenem, imipenem-cilastatin, and meropenem include:

- ✓ nausea and vomiting
- ✓ diarrhea.
- ✓ **Hypersensitivity reactions**, such as rashes, may occur, particularly in the patient with a known hypersensitivity to penicillins.
- ✓ **Kidney conditions** (In the patient with decreased or impaired renal function, the dosage may need to be adjusted).

Vancomycin

- Glycopeptide from *Streptomyces*
- Inhibition of cell wall synthesis
- Used to kill **MRSA**
- Emerging Vancomycin resistance: **VRE** and **VRSA**



Pharmacodynamics

- ✓ Vancomycin inhibits bacterial cell-wall synthesis,
- ✓ When the bacterial cell wall is damaged, the body's natural defenses can attack the organism.

Pharmacotherapeutics

Vancomycin is active against gram-positive organisms, such as *S. aureus*, *S. epidermidis*, *S. pyogenes*, *Enterococcus*, and *S. pneumoniae*.

In the I.V. league

I.V. vancomycin is the **therapy of choice** for the patient with a serious resistant staphylococcal infection who's hypersensitive to penicillins.

Pharmacokinetics

- ✓ Because vancomycin is **not absorbed** from the GI tract, it must be given I.V. to treat systemic infections.
- ✓ **an oral form of vancomycin** is used to treat pseudomembranous colitis.
- ✓ Vancomycin diffuses well into pleural, pericardial, synovial joint ascitic fluids.
- ✓ Normal **$T_{1/2}$** of vancomycin is **6 to 10 hr** , compared to over 200 hr in end-stage renal disease.

No switching!

Remember that I.V. vancomycin can't be used in place of oral vancomycin and vice versa. The two forms aren't interchangeable.

Metabolism and excretion

- ✓ The metabolism of vancomycin is minimal .
- ✓ About **90% of the dose** is excreted **unchanged in urine within 24 hours**.

Vancomycin Mupirocin

potent antibiotics often used in combination, particularly in topical formulations (e.g., 2% Mupirocin/5% Vancomycin gel) to treat severe, localized, or resistant skin infections like MRSA, impetigo, and folliculitis

Feature	Vancomycin	Mupirocin
Route	IV or oral	Topical (skin/nose)
Infection severity	Severe systemic	Local skin infections
Mechanism	Cell wall	Protein synthesis
MRSA	Yes	Yes (local)

Aztreonam

- a monocyclic beta-lactam antibiotic (a monobactam) originally isolated from *Chromobacterium violaceum*.
- Aztreonam exhibits potent and specific activity in vitro against a wide spectrum of gram-negative aerobic pathogens including *Pseudomonas aeruginosa*.

plasma (cell) membrane inhibitors.

Feature	Polymyxin	Gramicidin
Main target	Gram-negative	Gram-positive
Use	Topical ± systemic (last line)	Topical only
Mechanism	Plasma Membrane disruption	PlasmaMembrane ion channels
Toxicity	Kidney & nerve toxicity	Systemic toxicity → only topical

Antibiotics that inhibit the folate synthesis

- These drugs inhibit bacterial DNA/RNA synthesis by blocking the production of essential **folic acid**.

- a) **Trimethoprim**: Primarily used for **UTIs**; can cause **hyperkalemia** or blood disorders like megaloblastic anemia.

- b) **Sulfonamides**: Used for **Nocardiosis** and burns; linked to **crystalluria** and severe **Stevens-Johnson Syndrome**.

- Co-Trimoxazole**: A synergistic 1:5 combo (Trimethoprim/Sulfamethoxazole) used for **pneumonia**.

- Combining them creates a more powerful bactericidal effect than using either drug alone. High risk of **allergic reactions** and potential for **kernicterus** (brain damage) in newborns.

- Contraindications**: Avoid in **pregnancy**, nursing, infants under 2 months, or those with known **sulfa allergies**.